Low-Cost EOG-Based Human–Computer Interaction Using Machine Learning for Eye-Movement and Blink Control

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Abstract

Electrooculography (EOG) provides a non-invasive means of measuring electrical changes associated with eye movements and blinking, making it useful for hands-free human–computer interaction. This work presents the design and evaluation of a low-cost EOG-based interface that recognizes a small set of ocular actions and converts them into real-time computer commands. The acquisition system uses a BioAmp EXG Pill connected to an ESP32 microcontroller, with electrodes positioned above and below the right eye for vertical EOG recording. Signals are sampled at 500 Hz and processed through embedded baseline and frequency filtering before transmission to a computer.

The collected dataset contains 64,000 EOG samples organized into 160 trials, with 40 trials each for neutral gaze, upward eye movement, downward eye movement, and blinking. Each 0.8 s trial contains 400 samples. A 28-dimensional handcrafted representation was constructed from statistical, temporal, peak-related, frequency-domain, and Hjorth-mobility features. Random Forest, Gradient Boosting, and radial-basis-function Support Vector Machine classifiers were evaluated.

The original implementation produced 89.27% five-fold cross-validation accuracy and 91.14% hold-out accuracy after generating heavily overlapping windows. Because this evaluation can place strongly correlated windows in both training and test sets, an additional trial-level re-analysis was performed using the original 160 trials as independent evaluation units. Under five-fold stratified trial-level cross-validation, Random Forest achieved the strongest performance, with an accuracy of **78.75% ± 10.90%** and a macro F1-score of approximately **0.79**. Blink and neutral gaze were the most reliably recognized classes, whereas greater confusion occurred between upward and downward eye movements. Feature-group analysis showed that using all 28 features provided higher trial-level accuracy than using statistical, frequency, or temporal feature groups alone.

The selected EOG classes were also integrated into a real-time prototype in which upward and downward movements control vertical cursor motion and a double blink generates a mouse click. Confidence filtering, temporal voting, and an action cooldown are used to reduce unstable commands. The results demonstrate the feasibility of a low-cost machine-learning-based EOG interaction pipeline while also showing the importance of trial-independent evaluation. Participant-level metadata was not preserved in the available dataset, so subject-independent generalization cannot yet be assessed.

Keywords: electrooculography, EOG, human–computer interaction, eye movement classification, blink detection, machine learning, assistive interface, ESP32, biosignal processing

1. Introduction

1.1. Background

Human–computer interaction commonly depends on physical input devices such as keyboards, mice, touchscreens, joysticks, and game controllers. These interfaces are appropriate for most users, but there are situations in which direct manual interaction is difficult, undesirable, or impossible. Physiological signals provide an alternative interaction channel because measurable changes generated by the human body can be interpreted and translated into commands for external systems.

Brain–computer interface research has established a general signal-processing framework in which physiological activity is acquired, transformed into useful signal representations, classified, and translated into control commands [Wolpaw et al., 2002]. Electroencephalography (EEG) is commonly associated with BCI research, whereas electrooculography (EOG) captures electrical changes generated by ocular activity. EOG should therefore be distinguished from direct measurement of neural activity even when it is used alongside EEG in broader physiological-computing systems.

EOG is based on the electrical potential difference between the anterior and posterior regions of the eye. Changes in eye orientation modify the electrical potential measured by electrodes positioned around the eyes, producing signals that can be used to distinguish eye movements and blinking [Chang, 2019]. Reviews of EOG-based human–computer interaction have documented applications including pointing interfaces, virtual keyboards, wheelchairs, communication systems, gaming interfaces, and other hands-free control systems [Chang, 2019; Belkhiria et al., 2022].

Several properties make EOG attractive for small-scale interaction systems. It can be measured using relatively inexpensive electrodes and amplification hardware, does not require continuous optical visibility of the eye, and can generate comparatively pronounced responses during intentional eye movements and blinking. At the same time, EOG signals are affected by baseline drift, electrode placement, muscular interference, involuntary blinking, recording conditions, and variation between users and sessions [Chang, 2019; Belkhiria et al., 2022]. A practical EOG interface must therefore address both signal classification and the conversion of noisy predictions into stable application commands.

1.2 Literature Review

Existing EOG systems have approached this problem in different ways. Ang et al. [2015] demonstrated a wearable single-channel EOG interface in which double-blink patterns were translated into cursor operations. Martínez-Cerveró et al. [2020] developed an open software/hardware EOG platform for classifying directional eye movements using conventional signal processing and machine learning. More recently, Hernández Pérez et al. [2023] investigated wavelet-derived EOG representations with supervised algorithms including K-nearest neighbors, Support Vector Machines, and decision trees.

These studies demonstrate that EOG interaction is feasible, but their reported performance cannot be compared directly because electrode arrangements, hardware, participants, gesture definitions, preprocessing pipelines, datasets, and validation strategies differ considerably. Evaluation methodology is particularly important. When neighboring segments of a physiological recording overlap substantially, a

random window-level train/test split can place nearly identical signal content in both partitions and result in an optimistic estimate of generalization.

The present work investigates a comparatively simple question: whether an inexpensive vertical-EOG acquisition system can recognize four discrete ocular states—neutral gaze, upward movement, downward movement, and blinking—and use those states to support a basic real-time computer-control prototype.

The project was originally developed as a broader EEG/EOG human–computer interaction system containing game-control and electrical-device demonstrations. However, the available experimental data and reproducible machine-learning pipeline are substantially stronger for the EOG component. Accordingly, this paper focuses its quantitative evaluation on EOG classification and treats the EEG-based focus applications as prototype extensions rather than experimentally validated contributions.

The principal contributions of this work are:

1. a low-cost vertical EOG acquisition system based on a BioAmp EXG Pill and ESP32 microcontroller;
2. a balanced four-class dataset containing 160 recorded EOG trials and 64,000 signal samples;
3. a 28-dimensional handcrafted feature representation combining statistical, temporal, peak, spectral, and signal dynamic information;
4. comparison of Random Forest, Gradient Boosting, and RBF-SVM classifiers;
5. a trial-level re-analysis designed to reduce the leakage risk present in the original overlapping-window evaluation;
6. a feature-group ablation analysis using the Research gap and Motivationavailable experimental data; and
7. Integration of the EOG classifier with a real-time control layer incorporating confidence gating, temporal voting, cursor movement, and double-blink clicking.

The objective is not to claim a new EOG sensing principle or a new machine-learning algorithm. Instead, the contribution lies in the reproducible combination and evaluation of inexpensive acquisition hardware, engineered EOG features, conventional classifiers, and a real-time interaction pipeline.

2. Related Work

2.1 Electrooculography for Human–Computer Interaction

Electrooculography records electrical variations associated with movement of the eye. The eye can be approximated electrically as a dipole in which the corneal region is relatively positive with respect to the retinal region. Movement of this dipole changes the voltage recorded by electrodes around the eyes [Chang, 2019].

The placement of electrodes depends on the movement of interest. Electrodes positioned on opposite horizontal sides of the eyes are commonly used for horizontal movement, whereas electrodes positioned above and below an eye can capture vertical movement. Blinking also produces a characteristic transient in vertical EOG because movement of the eyelid and ocular structures substantially affects the measured potential.

Chang [2019] reviewed the use of EOG for HCI and described applications involving computer interaction, communication, mobility systems, and assistive interfaces. Belkhiria et al. [2022] subsequently reviewed EOG-based HCI work published between 2000 and 2020, organizing such systems around stages including acquisition, processing, translation algorithms, and interaction commands.

These observations motivate the architecture used in the present work: the physiological signal is first acquired and conditioned, transformed into quantitative descriptors, classified into a small action vocabulary, and finally passed through a decision layer before an operating-system command is executed.

2.2 EOG Cursor-Control Systems

Cursor interaction has been a recurring application of EOG because directional eye movements and blinking can naturally be mapped to navigation and selection operations.

Ang et al. [2015] proposed a wearable single-channel system based on double-blink EOG activity. Their implementation demonstrated that a small ocular command vocabulary can be used to control cursor navigation and actions without requiring conventional manual input.

Martínez-Cerveró et al. [2020] presented an open software/hardware platform for EOG interaction and evaluated classification of multiple directional eye movements. Their work is particularly relevant to the present project because it demonstrates that conventional feature extraction and supervised machine learning remain useful for practical EOG control without necessarily requiring deep neural architectures.

The system presented here differs in its specific hardware, signal representation, classes, and interaction logic. Direct numerical comparisons are therefore avoided.

2.3 Machine Learning for EOG Classification

A range of supervised classifiers has been applied to EOG signals. Hernández Pérez et al. [2023] evaluated EOG movement classification using wavelet-derived features and algorithms including K-nearest neighbors, Support Vector Machines, and decision trees.

The present study uses three conventional classifiers: Random Forest, Gradient Boosting, and an RBF Support Vector Machine.

Random Forest combines multiple randomized decision trees and aggregates their predictions [Breiman, 2001]. It can represent nonlinear decision boundaries and interactions between engineered signal features.

Gradient Boosting constructs a sequence of weak predictive models in which later models attempt to reduce errors made by the existing ensemble [Friedman, 2001].

Support Vector Machines construct decision boundaries that maximize class separation in an appropriate feature space. Kernel functions allow nonlinear class boundaries to be represented without explicitly constructing the full transformed feature space [Cortes and Vapnik, 1995].

These models were selected because they are suitable for moderate-dimensional structured feature vectors and can perform inference without the computational requirements associated with larger deep-learning architectures.

2.4 EOG Feature Engineering

Different ocular actions affect multiple properties of the waveform. A purely amplitude-based representation can therefore discard potentially useful temporal or spectral information.

The present system combines summary statistics, waveform derivatives, slope, zero crossings, peak characteristics, temporal segment means, frequency-domain energy, dominant frequency, and Hjorth mobility.

Hjorth [1970] introduced time-domain descriptors for electrophysiological signals, including mobility, which relates variation in the derivative of a signal to variation in the original waveform. In this work, Hjorth mobility is used only as a mathematical signal descriptor and is not interpreted as a physiological biomarker.

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3. Research Problem and Objectives

The problem considered in this work is the recognition of four ocular states from short vertical-EOG recordings.
For an EOG window x, a feature-extraction function produces

$$\mathbf{z} = \phi(x) \in \mathbb{R}^{28},$$

where ϕ represents the 28-dimensional handcrafted feature transformation.

A classifier then estimates

$$f(\mathbf{z}) = \hat{y},$$

where

$$\hat{y} \in \{\text{Neutral}, \text{Up}, \text{Down}, \text{Blink}\}.$$

Three research questions are addressed.

RQ1

How accurately can the four ocular states be distinguished when complete recorded trials, rather than highly overlapping windows, are treated as independent evaluation units?

RQ2

Among Random Forest, Gradient Boosting, and RBF-SVM, which conventional classifier performs most consistently under trial-level cross-validation?

RQ3

Do the statistical, temporal, and spectral feature groups provide complementary information for four-class EOG recognition?

A fourth engineering question concerns real-time integration:

RQ4

Can the trained classification pipeline be integrated with a decision layer capable of generating vertical cursor movement and blink-based clicking in real time?

RQ4 is evaluated at the implementation level rather than through a controlled usability study; no quantitative end-to-end latency or task-completion claims are made.

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4. System Architecture

The implemented processing chain can be summarized as

Eye activity →Electrodes →BioAmp EXG Pill →ESP32 →Serial Stream →Signal Processing →Feature Extra

4.1 EOG Acquisition Hardware

The bio-signal acquisition front end consists of a BioAmp EXG Pill connected to an ESP32 development board.

The analog output of the BioAmp module is connected to GPIO 34 of the ESP32. The microcontroller is configured for 12-bit analog-to-digital conversion.

The EOG electrode arrangement used in the supplied implementation is:

- **IN+**: above the right eye at the supraorbital ridge;
- **IN−**: below the right eye at the infraorbital ridge;
- **REF**: earlobe or mastoid region.

This is a vertical EOG configuration. Consequently, the experimentally supported directional classes are UP and DOWN. Horizontal LEFT and RIGHT cursor commands described in an earlier project synopsis are not included in the present study because the supplied dataset and trained classifier do not contain these classes.

4.2 Sampling

The microcontroller is configured to acquire data at

$$f_s = 500 \text{ Hz}.$$

The nominal sampling interval is therefore

$$T_s = \frac{1}{500} = 0.002 \text{ s}.$$

A single 0.8 s trial nominally contains

$$500 \times 0.8 = 400$$

samples.

Communication with the computer is performed at 115200 baud.

The ESP32 transmits records in the form

$$(\text{timestamp}_{\mu s}, \text{raw ADC}, \text{filtered voltage}).$$

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5. Embedded Signal Preprocessing

Initial preprocessing is performed before the signal reaches the computer.

5.1 Voltage Conversion

The 12-bit ADC generates values in the range 0–4095. The firmware converts the ADC reading to a signal centered around the BioAmp mid-rail voltage:

$$V[n] = \frac{ADC[n]}{4095}(3.3) - 1.65.$$

5.2 High-Pass Filtering

A first-order high-pass stage is implemented to reduce slow drift and DC components. The implemented recurrence is

$$y_{HP}[n] = \alpha_{HP}\left(y_{HP}[n-1] + x[n] - x[n-1]\right),$$

with

$$\alpha_{HP} \approx 0.99373.$$

The nominal cutoff used when deriving this coefficient is approximately 0.5 Hz.

5.3 Low-Pass Filtering

A first-order low-pass stage is subsequently applied:

$$y_{LP}[n] = \alpha_{LP} x[n] + (1-\alpha_{LP}) y_{LP}[n-1],$$

where

$$\alpha_{LP} \approx 0.27363.$$

The nominal cutoff is approximately 30 Hz.

5.4 Moving Baseline Correction

The firmware maintains a 500-sample moving average:

$$B[n] = \frac{1}{500}\sum_{k=n-499}^{n} y_{LP}[k].$$

The transmitted signal is approximately

$$x_{out}[n] = y_{LP}[n] - B[n].$$

At 500 Hz, the 500-sample baseline corresponds to approximately one second of signal history.

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6. Data Collection

6.1 Ocular Classes

The acquisition procedure contains four labeled actions.

Label ID	Class	Recording instruction
1	Neutral	Look straight ahead and keep the eyes still
2	Up	Perform a deliberate upward eye movement
3	Down	Perform a deliberate downward eye movement
4	Blink	Perform one natural blink

Each trial is preceded by a settling period. The classes are scheduled in randomized order rather than recording all trials of one class sequentially.

6.2 Number of Trials

The collection configuration specifies 40 trials per class.

Therefore,

$$40 \times 4 = 160$$

trials were collected.

Each trial contains 400 samples, resulting in

$$160 \times 400 = 64,000$$

rows in the supplied dataset.

The class distribution is exactly balanced:

Class	Trials	Signal sample
Neutral	40	16000
Up	40	16000
Down	40	16000
Blink	40	16000
Total	**160**	**64,000**

The stored dataset fields are:

- timestamp_us ,
- raw_adc ,

• filtered_v ,
• label_id ,
• label_name .

6.3 Reconstruction of Trial Boundaries

The original CSV does not contain an explicit **trial_id** field.

However, the acquisition software records exactly 400 samples for each trial, and the supplied 64,000-row dataset consists of 160 consecutive 400-row blocks. Inspection of these blocks showed that every block contains a single consistent class label.

Consequently, for the additional publication-oriented analysis presented in this paper, the dataset was reconstructed as 160 independent acquisition trials:

$$T_i = \{x_{400i}, \ldots, x_{400i+399}\}.$$

This reconstruction allows trial-level evaluation without collecting or fabricating new experimental observations.

Participant identity cannot be reconstructed from the available files. Subject-independent conclusions are therefore not made.

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7. Computer-Side Preprocessing and Feature Extraction

The original training pipeline applies an additional 0.5–35 Hz band-pass filter followed by a 50 Hz notch filter to the signal available on the computer.

For each analysis window, 28 features are extracted.

7.1 Statistical Features

The first ten features summarize the signal distribution and amplitude characteristics:.

1. mean,
2. standard deviation,
3. minimum,
4. maximum,
5. peak-to-peak range,
6. root mean square,
7. skewness,
8. kurtosis,
9. sum of absolute amplitudes,
10. mean absolute first derivative.

The arithmetic mean is

$$\mu = \frac{1}{N}\sum_{n=1}^{N} x_n.$$

Standard deviation is

$$\sigma = \sqrt{\frac{1}{N}\sum_{n=1}^{N}(x_n - \mu)^2}.$$

Peak-to-peak range is

$$R = \max(x) - \min(x).$$

RMS is

$$RMS = \sqrt{\frac{1}{N}\sum_{n=1}^{N} x_n^2}.$$

The mean absolute derivative is

$$MAD_\Delta = \frac{1}{N-1}\sum_{n=2}^{N} |x_n - x_{n-1}|.$$

7.2 Trend and Zero-Crossing Features

A linear regression slope is calculated across the samples of each window.

The zero-crossing rate is

$$ZCR = \frac{\sum_{n=2}^{N} I[\text{sign}(x_n) \; / \qquad}{N}. \qquad = \text{s}$$

7.3 Peak Features

Positive and negative peaks are detected using a fixed prominence condition.

Four peak-related variables are extracted:

- number of positive peaks,
- number of negative peaks,
- maximum positive peak,
- minimum negative peak.

7.4 Temporal Segment Features

Each window is divided into approximately three equal temporal segments.

The mean of each segment is computed:

$$\mu_1, \quad \mu_2, \quad \mu_3.$$

A final temporal-trend variable is

$$D_{31} = \mu_3 - \mu_1.$$

These features represent how the waveform changes from the beginning to the end of the trial.

7.5 Frequency-Domain Features

The discrete Fourier representation is calculated using a real-valued FFT.

Energy-related features are derived for:

- 0–2 Hz,
- 2–8 Hz,
- 8–15 Hz,
- 15–30 Hz.
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For frequency interval $[f_l, f_h)$, the implemented band-power descriptor has the form

$$P(f_l, f_h) = \frac{\sum_{k \in \mathcal{F}(f_l, f_h)} |X[k]|^2}{N f_s}.$$

Relative 0–2 Hz and 2–8 Hz features are calculated in addition to their absolute values.

The dominant non-DC frequency is also included.

These frequency intervals are used as engineering features. They are not interpreted in this paper as EEG cognitive frequency bands because the recorded signal is EOG.

7.6 Hjorth Mobility

The 28th feature is Hjorth mobility [Hjorth, 1970]:

$$M = \sqrt{\frac{Var(\Delta x)}{Var(x) + \epsilon}}.$$

Here, Δx is the first difference of the signal and ϵ is a small numerical constant.

8. Machine-Learning Models

8.1 Random Forest

The Random Forest configuration uses:

- 300 estimators,
- maximum tree depth of 15,
- minimum leaf size of 2,
- balanced class weighting,
- random seed 42.

Random Forest was included because it can capture nonlinear interactions among heterogeneous engineered features [Breiman, 2001].

8.2 Gradient Boosting

The Gradient Boosting configuration uses:

- 200 estimators,
- learning rate 0.08,
- maximum depth 5,
- subsampling ratio 0.8,
- random seed 42.

The model progressively constructs an additive ensemble of trees [Friedman, 2001].

8.3 RBF Support Vector Machine

The Support Vector Machine uses:
- radial-basis-function kernel,
- $C = 10$
- gamma="scale" ,
- balanced class weighting,
-
probability estimation enabled.

SVMs provide nonlinear classification through kernel-based feature-space transformations [Cortes and Vapnik, 1995].

8.4 Standardization

Each classifier is placed inside a pipeline containing a standard-scaling stage:

$$z_j = \frac{x_j - \mu_j}{\sigma_j}.$$

The same transformation is fitted using training data within each evaluation fold.

9. Evaluation Methodology

Two forms of evaluation are distinguished because they answer different questions.

9.1 Original Window-Level Evaluation

The original training implementation used a 400-sample window and a 50-sample step.

Thus, consecutive windows had

$$400 - 50 = 350$$

shared samples.

The overlap percentage was

$$\frac{350}{400} \times 100 = 87.5\%$$

The original pipeline created these windows before performing model evaluation.

Using this procedure, Gradient Boosting was selected as the best model with:

- five-fold cross-validation accuracy: **89.27%**;
- 80/20 hold-out accuracy: **91.14%**.

The reported hold-out classification metrics were:

Class	Precision	Recall	F1	Support
Neutral	1.00	0.86	0.93	22
Up	0.81	0.94	0.87	18
Down	0.89	0.85	0.87	20
Blink	0.95	1.00	0.87	19

The corresponding confusion matrix was

$$[19\ 1\ 1\ 10\ 17\ 1\ 0\ 0\ 3\ 17\ 0\ 0\ 0\ 0\ 19].$$

These results describe the original system but are not treated as the primary evidence of generalization in this paper because windows derived from the same recordings can be strongly correlated.

9.2 Trial-Level Evaluation

To provide a stricter analysis without collecting additional data, the original 160 acquisition trials were reconstructed and each entire 400-sample trial was treated as one independent example.

Thus,

$$N_{evaluation} = 160,$$

with 40 examples per class.

Five-fold StratifiedKFold cross-validation was used with shuffling and random seed 42.
Each fold therefore contained complete trials rather than partially overlapping fragments of another evaluation example.

The same 28 features and original model hyperparameters were used.

Accuracy and macro F1 were calculated.

For classification,

$$Accuracy = \frac{\text{Correct predictions}}{\text{Total predictions}}.$$

For a particular class,

$$Precision = \frac{TP}{TP + FP},$$

$$Recall = \frac{TP}{TP + FN},$$

and

$$F1 = \frac{2PR}{P + R}.$$

Macro F1 gives equal importance to all four classes:

$$F1_{macro} = \frac{1}{4}\sum_{c=1}^{4} F1_c.$$

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10. Results

10.1 Trial-Level Model Comparison

The stricter trial-level evaluation produced lower performance than the original overlapping-window analysis.

Table 1. Five-fold trial-level classification performance

Classifier	Accuracy, mean ± SD	Macro F1, mean ± SD
Random Forest	**78.75% ± 10.90%**	**78.68% ± 10.80%**
Gradient Boosting	73.75% ± 3.19%	73.82% ± 3.76%
RBF-SVM	71.25% ± 6.67%	70.70% ± 6.91%

Random Forest achieved the highest mean trial-level accuracy.

This differs from the original window-level analysis, in which Gradient Boosting was selected as the strongest classifier.

The difference is significant for interpretation: the identity of the best-performing model depends on the evaluation unit. The result illustrates why classification studies using overlapping physiological windows should report carefully how training and testing samples are separated.

10.2 Trial-Level Random Forest Performance

Pooling predictions from the five trial-level cross-validation folds produced:

- accuracy: **0.7875**;
- macro precision: **0.7848**;
- macro recall: **0.7875**;
- macro F1: **0.7845**.

Table 2. Class-specific trial-level performance

Class	Precision	Recall	F1	Support
Neutral	0.80	0.90	0.85	40
Up	0.78	0.73	0.75	40
Down	0.72	0.65	0.68	40
Blink	0.83	0.88	0.85	40
Macro average	**0.78**	**0.79**	**0.78**	**160**

Neutral gaze and blinking were the most reliably classified states under the stricter evaluation.

Downward movement produced the lowest recall, at 0.65.

10.3 Trial-Level Confusion Matrix

The pooled confusion matrix was

[36 1 1 22 29 7 26 5 26 31 2 2 35],

where rows represent true classes and columns represent predicted classes in the order Neutral, Up, Down, Blink.

Table 3. Trial-level confusion matrix

Actual Predicted	Neutral	Up	Down	Blink
Neutral	36	1	1	2
Up	2	29	7	2
Down	6	5	26	3
Blink	1	2	2	35

The largest pairwise confusion occurred between upward and downward movements.

Seven upward trials were classified as downward, while five downward trials were classified as upward.

This suggests that directional vertical movements are less separable under trial-independent evaluation than the original overlapping-window result implied.

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11. Feature-Group Analysis

The complete model contains 28 features. To examine whether the different feature families contribute complementary information, an additional ablation analysis was performed using Random Forest and the same five-fold trial-level evaluation.

No new experimental observations were introduced.

The following feature groups were evaluated:

1. statistical features;
2. trend, peak, and temporal-segment features;

3. frequency-domain features;
4. all time-domain features together with Hjorth mobility;
5. all 28 features.

Table 4. Feature-group ablation results

Feature set	Trial-level accuracy, mean ± SD	Macro F1
Statistical features	76.25% ± 11.46%	0.760
Trend + peak + segment features	68.13% ± 9.56%	0.674
Frequency-domain features	75.63% ± 4.59%	0.753
Time-domain + Hjorth	75.00% ± 7.65%	0.748
All 28 features	**78.75% ± 10.90%**	**0.785**

No individual feature group exceeded the complete representation.

The difference is moderate rather than dramatic, so the results do not establish that every individual feature is necessary. They do, however, indicate that the complete representation performed better than any of the tested broad feature subsets on this dataset.

Statistical features alone reached 76.25%, and frequency-domain features reached 75.63%, suggesting that both waveform amplitude/distribution and spectral structure carry useful class information.

The trend/peak/segment group performed least strongly when used on its own.

A more detailed feature-selection study would require a larger dataset before strong conclusions could be drawn.

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12. Real-Time Control Implementation

The trained machine-learning pipeline was integrated into a Python application for real-time EOG interaction.

12.1 Streaming Window

Incoming EOG values are stored in a rolling 400-sample buffer.

Inference is requested every 50 samples.

At 500 Hz, the prediction update interval is nominally

$$\frac{50}{500} = 0.1 \text{ s}.$$

This number represents how often the algorithm can update once a complete buffer is available. It should not be interpreted as measured end-to-end command latency.

12.2 Classification Confidence

For each window, the classifier provides class probabilities.

A command is considered only if the predicted-class probability satisfies

$$p(\hat{y}|x) \geq 0.55.$$

Otherwise, the prediction is marked as low confidence and does not generate an action.

12.3 Temporal Voting

A buffer containing the four most recent class predictions is maintained.

The controller uses the most frequent label:

$$\hat{y}_{stable} = mode(\hat{y}_{t-3}, \hat{y}_{t-2}, \hat{y}_{t-1}, \hat{y}_t).$$

This provides simple temporal smoothing and reduces the influence of a single unstable classification.

12.4 Action Cooldown

An action cooldown of

$$0.25 \text{ s}$$

is applied to prevent rapid repeated cursor actions.

12.5 Cursor Mapping

The evaluated EOG classes are mapped to computer actions as follows:

Classified state	Action

Neutral	No movement
Up	Cursor moves upward
Down	Cursor moves downward
Blink	Blink-event state updated

The default vertical displacement is 30 pixels per accepted directional action.

12.6 Double-Blink Click

A single recognized blink is stored as a temporary event.

If another blink is detected within

$$700 \text{ ms},$$

the controller executes a mouse click.

Thus,

$$Blink_1 + Blink_2 \quad \text{with} \quad \Delta t < 700 \text{ ms}$$

produces the click action.

The system therefore separates the machine-learning problem—recognition of individual EOG classes— from the interaction problem of determining which sequence of predictions should produce an operating- system command.

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13. Broader EEG/EOG Prototype

The broader engineering project from which this work was developed included additional demonstrations:

- EOG blink interaction with a browser game;
- EEG-derived focus/relaxation control for a racing game;
- focus-dependent control of an electrical light through a relay;
- EOG-based computer cursor interaction.

The supplied focus-control application receives a numerical value described as Focus Level (%) and maps that value to keyboard commands according to a threshold.

However, the available implementation does not contain the algorithm by which raw EEG is transformed into this focus value.

Accordingly, the present paper does **not** claim:

- measured EEG-focus classification accuracy;
- validated attention estimation;
- quantitative racing-game performance;
- quantitative EEG-controlled appliance performance.

These components are retained as broader prototype demonstrations but are not included in the experimentally evaluated contribution of the paper.

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14. Discussion

The results provide two different views of the EOG classifier.

The original window-level experiment produced a hold-out accuracy of 91.14%, while the stricter trial-level analysis produced a mean cross-validation accuracy of 78.75% with Random Forest.

This difference is important.

It would be tempting to report only the larger number. However, doing so would give an incomplete description of the system's generalization because the original windows overlapped by 87.5%.

For two consecutive windows,

$$W_1 = x[1 : 400]$$

and

$$W_2 = x[51 : 450],$$

350 samples are shared.

If W_1 enters training and W_2 enters testing, the test example contains much of the same underlying physiological recording as the training example.

The reconstructed trial-level experiment removes this particular form of overlap between evaluation examples. Its lower accuracy is therefore a more conservative and scientifically useful indication of how the classifier performs on complete recorded trials not duplicated through overlapping windows.

The model-ranking difference is also notable. Gradient Boosting was strongest in the original analysis, whereas Random Forest was strongest at the trial level. The objective of this work is not to argue that Random Forest is universally superior for EOG.

Instead, the result demonstrates that classifier ranking can depend on how the dataset is partitioned.

The trial-level confusion matrix also provides a more realistic picture of class difficulty.

Neutral trials were identified with 90% recall, and blink trials with 87.5% recall. Up and Down were more frequently confused.

This is consistent with the fact that both directional classes originate from vertical eye movement measured using the same electrode pair. However, the current experiment cannot determine precisely whether the errors arise from physiological similarity, electrode placement, movement timing, subject variation, preprocessing, or the feature representation. Such explanations should therefore be considered hypotheses rather than conclusions.

The feature-group experiment provides additional evidence regarding the representation. Statistical features alone produced 76.25% accuracy and frequency-domain features 75.63%, while the full feature vector reached 78.75%. The difference is not sufficiently large to claim that the 28-feature design is optimal, but it indicates that combining heterogeneous signal characteristics is at least beneficial for this dataset.

The real-time architecture also illustrates that classification accuracy alone is not equivalent to interaction quality.

An interface may have reasonable average classification accuracy but still feel unstable if isolated false predictions immediately trigger actions. For this reason, the implementation uses a probability threshold, majority voting over four predictions, and an action cooldown.

These design choices improve command stability at the cost of additional responsiveness constraints. Since actual end-to-end command latency was not measured, no numerical latency claims are made.
Compared with prior EOG cursor systems [Ang et al., 2015; Martínez-Cerveró et al., 2020], the present system should be viewed as a low-cost prototype and experimental platform rather than as a direct performance competitor. Differences in hardware, classes, participant populations, tasks, and evaluation protocols prevent meaningful comparison of headline accuracy values alone.

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15. Limitations

15.1 Participant Metadata Is Unavailable

The most important remaining limitation is the absence of participant identifiers in the stored dataset.

The current files establish that 160 trials were recorded, but they do not establish how many unique participants produced those trials.

As a result, this paper does not claim subject-independent performance.

A future dataset should include at minimum:

participant_id, *trial_id*, *session_id*, *class*, *timestamp*, *raw_signal*, *filtered_signal*.

15.2 Trial IDs Were Not Originally Stored

Trial IDs were reconstructed from the known 400-sample acquisition protocol.

This is sufficient for the present dataset because all 160 consecutive blocks contain exactly 400 samples and consistent labels, but explicit trial IDs should be recorded in future experiments.

15.3 Limited Number of Trials

Only 40 trials per class are available.

Although this is sufficient for preliminary comparison of conventional models, it is too small to establish robust population-level performance.

The comparatively large fold-to-fold variation of Random Forest accuracy

$$78.75\% \pm 10.90\%$$

also indicates that performance estimates remain sensitive to which trials are assigned to individual folds.

15.4 No Subject-Independent Evaluation

Leave-one-subject-out validation cannot be performed because participant identity was not preserved.
This prevents assessment of whether a model trained on one user would generalize to another without recalibration.

15.5 Restricted Command Set

The implemented classifier contains only:

- Neutral,
- Up,
- Down,
- Blink.

LEFT and RIGHT eye movements are not present in the supplied dataset or trained classifier.

Accordingly, this paper does not claim four-directional cursor control.

15.6 Offline and Real-Time Filtering Differ

The original offline training implementation uses forward-backward filtering, whereas real-time processing necessarily uses causal streaming filters.

Forward-backward filtering removes phase shift by using future as well as past signal information and therefore cannot be reproduced exactly during live operation.

Future training should use the same causal filtering architecture used during deployment.

15.7 End-to-End Interaction Latency Was Not Measured

Inference is scheduled every 100 ms after the window buffer is populated, but this is not equivalent to command latency.

Actual latency would include:

- acquisition,
- buffering,
- filtering,
- feature extraction,
- model inference,
- prediction smoothing,
- cooldown effects,
- operating-system input execution.

No measured mean, median, or percentile latency is available.

15.8 No Controlled HCI User Study

The interface was implemented and demonstrated, but no structured study measured:

- command success rate,
- false activation frequency,
- missed command rate,
- target-selection accuracy,
- double-blink success rate,
- task completion time,
- user fatigue.

Consequently, claims are restricted to classification and system implementation rather than usability.

15.9 EEG Focus Component Cannot Be Quantitatively Evaluated

The broader project includes a focus-control interface, but the underlying focus-estimation algorithm and associated raw experimental dataset were not available.

No accuracy or physiological interpretation of the focus subsystem is therefore reported.

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16. Future Work

The most useful next step is not to add additional machine-learning algorithms but to improve the experimental design.

16.1 Multi-Participant Data Collection

Future recording software should save participant, session, and trial identifiers.

A dataset collected from multiple participants would allow both user-dependent and user-independent evaluation.

16.2 Leave-One-Subject-Out Evaluation

With participants, leave-one-subject-out cross-validation could train on

$$S - 1$$

participants and evaluate on the remaining person.

The procedure would be repeated S times.
This would answer a substantially stronger question than random window classification: whether the EOG representation generalizes to a person whose signals were not seen during training.

16.3 Consistent Offline and Online Filtering

The real-time causal preprocessing chain should be reused during training.

A single pipeline would reduce mismatch between the waveforms encountered during model fitting and deployment.

16.4 Horizontal EOG

Additional electrodes positioned for horizontal EOG would allow explicit recording of LEFT and RIGHT movements.

A future class vocabulary could therefore contain:

$$\{Neutral, Up, Down, Left, Right, Blink\}.$$

This should only be reported once an actual horizontal-EOG dataset has been collected.

16.5 Real-Time HCI Evaluation

A controlled experiment could ask participants to perform fixed tasks such as:

1. move upward to a target;
2. move downward to a target;
3. execute a double-blink click;
4. perform a predefined sequence of directional and click commands.

Useful metrics would include:

- command-level accuracy;
- false activations per minute;
- missed commands;
- click success;
- task completion time;
- end-to-end response latency.

16.6 Feature Selection

The present ablation experiment suggests that multiple feature groups contribute useful information.

With a larger dataset, future work could investigate systematic feature-selection methods to determine whether similar performance can be obtained with fewer than 28 features.
Reducing the representation could decrease computational cost and simplify embedded deployment.

16.7 EEG Component

If the broader EEG functionality is to become part of the research contribution, future work should retain:

- raw EEG recordings;
- electrode locations;
- precise filter definitions;
- the mathematical formulation of the focus metric;
- threshold-selection procedure;
- labeled experimental conditions;
- participant identifiers;

performance measurements.

Without these records, focus-related demonstrations should remain outside the quantitative claims.

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17. Conclusion

This paper presented the design and evaluation of a low-cost EOG-based human–computer interaction system for recognizing neutral gaze, upward movement, downward movement, and blinking.

Vertical EOG signals were acquired using a BioAmp EXG Pill and ESP32 microcontroller at a configured sampling rate of 500 Hz. The available dataset contains 64,000 samples organized into 160 balanced trials, with 40 trials per class.

Each trial was represented using 28 handcrafted features covering statistical amplitude characteristics, temporal trends, peak structure, frequency-domain information, and Hjorth mobility.

The original project pipeline reported 89.27% cross-validation accuracy and 91.14% window-level hold-out accuracy using Gradient Boosting. Because the original windows overlapped by 87.5%, the present work additionally reconstructed the underlying acquisition trials and performed trial-level evaluation.

Under five-fold stratified trial-level cross-validation, Random Forest achieved the highest mean accuracy:

$$78.75\% \pm 10.90\%$$

with mean macro F1:

$$78.68\% \pm 10.80\%$$

The trial-level analysis showed particularly strong recognition of neutral gaze and blinking, while upward and downward movements were more frequently confused.

Feature-group analysis produced its highest accuracy when all 28 features were used, although statistical and frequency-domain subsets independently retained substantial discriminatory information.

The classification system was also incorporated into a real-time interaction pipeline. Accepted upward and downward predictions control vertical cursor motion, while two detected blinks within 700 ms trigger a mouse click. Confidence filtering, four-prediction temporal voting, and a 250 ms action cooldown provide additional command stabilization.

The work demonstrates that a relatively inexpensive EOG acquisition setup can support a functioning machine-learning-based interaction prototype. More importantly, the comparison between overlapping- window and trial-level evaluation demonstrates the effect that experimental methodology can have on apparent classifier performance.

The current evidence should therefore be viewed as a preliminary single-dataset evaluation rather than proof of subject-independent

assistive performance. A multi-participant dataset with explicit participant and trial identifiers, consistent causal preprocessing, subject-independent validation, and quantitative interaction testing is required before stronger claims regarding generalization and usability can be made.

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